Nathan Labiosa

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Objective

Machine learning engineer and researcher with 3+ years of experience across deep learning, NLP, CV, RL, and multimodal models, seeking applied AI roles. Passion developed through first-author work in drug discovery, NeurIPS-accepted research on video robustness, research in embodied AI and generalization, and autonomous decision systems for space, with a focus on empirically grounded, interpretable ML.

Skills

Languages: Python, C++, SQL, Bash, Cypher (Neo4j)
ML Frameworks: PyTorch, TensorFlow, Hugging Face, LangChain, Scikit-learn, NumPy, Pandas
Infrastructure & Tools: AWS, Google Cloud Platform (GCP), Git, Docker, Linux
Concepts: NLP, Computer Vision, Reinforcement Learning, RAG, Graph Databases, MLOps

Education

University of Southern California — Los Angeles, CA
Master of Science - Computer Science — Sep 2025 - May 2027

University of Wisconsin - Madison — Madison, WI
Bachelor of Science - Biomedical Engineering and Computer Science — May 2025

Experience

Advanced Space — Westminster, CO
Machine Learning Intern — May 2025 - Aug 2025

- Built an automated ETL pipeline using Python and Neo4j to ingest and structure 5,000+ technical documents, reducing data retrieval latency by 30%
- Developed a GraphRAG system by building a modular MCP tool that integrates Cypher query generation; outperformed baseline vector search accuracy by 18% on complex, multi-hop queries.
- Created and optimized RL/IRL algorithms for multi-object simulation environments, achieving >85% success rates in latent goal inference through hyperparameter tuning and simulation testing.
- Contributed technical architecture and feasibility analysis for a NASA NIAC Phase 1 proposal, defining ML system requirements for autonomous space exploration.

Revilico Inc — Los Angeles, CA
Machine Learning Engineer — Apr 2024 - May 2025

- Architected and managed model training pipelines on AWS, optimizing compute resources to validate eight novel drug activity predictors.
- Standardized model inference scripts and artifacts, streamlining the handoff to DevOps and enabling seamless integration of models into the user-facing frontend.
- Engineered feature extraction tools for SMILES sequences, boosting IC50/EC50 prediction R^2 scores by ~0.12 through improved data representation.
- Designed deep learning architectures for proteomic classification, improving accuracy by 15-25% over industry baselines across 30 protein families.
- Built RL frameworks for de novo molecular generation, yielding 100+ novel compounds optimized for electronegativity, folding patterns, and pH balance.

Research Experience

University of Southern California — Los Angeles, CA
Research Assistant in Physical Superintelligence Lab — August 2025 - Present

- Developing meta-learning algorithms for Vision-Language-Action (VLA) models, enabling rapid adaptation to unseen physical dynamics (mass, friction)
- Engineering generalizable control policies capable of zero-shot transfer across varying robot kinematics and visual domains, bridging the sim-to-real gap

University of Wisconsin - Madison Madison, WI
Computer Vision Researcher May 2024 - May 2025
- Contributed to a novel framework for temporally stable and corruption-robust video inference, integrating stabilization adapters into frozen vision backbones.
- Headed design and evaluation of adversarial robustness experiments, delivering +11.8% accuracy under iterative attacks without retraining base networks.
- Crafted stabilizer modules for ResNet architectures, reducing fine-tuning time by 40% while maintaining state-of-the-art robustness.
- Constructed and benchmarked a binary human/nonhuman classification task on the DAVIS dataset (3,455 annotated frames) with a fine-tuned ResNet-50, optimizing training schedules and augmentation protocols.

University of Central Florida Orlando, FL
Visual Large Language Model Researcher May 2023 - Oct 2023
- Oversaw collaboration with researchers from the University of Central Florida and Meta, driving significant advancements in large language model research.
- Refined complex, 65 billion parameter AI models into streamlined, 7 billion parameter university-level models, reducing compute cost by 92% and boosting widespread research applicability.
- Devised a visual large language model (VLLM) training method using novel techniques, merging over 150 thousand images and 160 thousand text examples during multiple training stages.
- Attained 95% of industry-leading text-based models leveraging a practical sized VLLM.
- Utilized Pytorch and DeepSpeed frameworks to execute parallel training in a high-performance GPU cluster

University of Wisconsin - Madison Madison, WI
Quantitative Cell Imaging Researcher Feb 2023 - Sep 2023
- Constructed novel software tools for tumor micro-environment quantification employing deep learning in the LOCI research laboratory, attaining 95% IOU vs state-of-the-art models.
- Improved cell recognition throughput by 1.2x, enabling lab imaging pipelines to analyze ~10,000 cells/day.

Publications

Instant Video Models: Universal Adapters for Stabilizing Image-Based Networks
Dutson M., **Labiosa, N.**, Li, Y., and Gupta, M.
NeurIPS, 2025. doi: 10.48550/arXiv.2512.03014

Hybrid Transformer Model for Protein Classification
Labiosa, N., A. Kohli, C. Chung, and C. Korban.
bioRxiv, 2024. doi: 10.1101/2024.10.31.621421

Visual Information and Large Language Models: A Deeper Analysis
Labiosa, N., Huynh, D., Lim, S.
2023 UCF CRCV Catalog